



kubectl apply in Kubernetes

Introduction

kubectl apply is a **Kubernetes command** used to create or modify resources using YAML or JSON configuration files.

- What does **kubectl apply** do?
- Why is **kubectl apply** preferred over **kubectl create**?
- How can we use it to manage Kubernetes workloads?

This guide will help you **understand and use kubectl apply** effectively.

Section 1: Learn

What is **kubectl apply**?

kubectl apply is used to **create, update, and manage Kubernetes resources** declaratively.

Feature	Description
Declarative Resource Management	Allows defining resources in YAML files
Supports Updates	Modifies resources without deleting and recreating them
Efficient Scaling	Helps scale applications smoothly
Preserves Configuration	Keeps changes made to running resources

Why Use **kubectl apply**?

1. **Supports Infrastructure as Code** – Helps manage resources through YAML configurations.
2. **Enables Continuous Deployment** – Ideal for GitOps and automated



pipelines.

3. **Allows Incremental Changes** – Updates only the necessary configurations.

4. **More Reliable than `kubectl create`** – Ensures minimal downtime when updating applications.

Interesting Fact

Unlike `kubectl create`, which only works for new resources, `kubectl apply` is **idempotent**, meaning it applies changes without duplication.

Section 2: Practice

1. Basic `kubectl apply` Commands

Apply a YAML Configuration File

```
kubectl apply -f deployment.yaml
```

Apply Multiple Files at Once

```
kubectl apply -f deployment.yaml -f service.yaml
```

Apply All YAML Files in a Directory

```
kubectl apply -f ./manifests/
```

2. Updating Resources Using `kubectl apply`

Modify an Existing Deployment

- Edit the **YAML file** (e.g., change the replica count).



- Reapply the configuration:

```
kubectl apply -f deployment.yaml
```

- Verify changes:

```
kubectl get deployments
```

3. Checking the Status of Applied Resources

Check the Status of a Deployment

```
kubectl rollout status deployment my-app
```

Check the Status of a Pod

```
kubectl get pods
```

Describe a Resource in Detail

```
kubectl describe pod <pod-name>
```

4. Rolling Back Changes

View the History of a Deployment

```
kubectl rollout history deployment my-app
```

Undo the Last Applied Change

```
kubectl rollout undo deployment my-app
```



Section 3: Know More

Frequently Asked Questions

1. What is the difference between **kubectl create** and **kubectl apply**?
 - **kubectl create** is for **one-time resource creation**, while **kubectl apply** is for **managing and updating resources**.
2. Can **kubectl apply** update running resources?
 - Yes, it **updates only the modified fields** without restarting unchanged components.
3. How do I check the effect of **kubectl apply** before running it?
 - Use **kubectl diff -f <file>.yaml** to preview changes.
4. What happens if I apply a YAML file with a mistake?
 - Kubernetes rejects invalid configurations, and no changes are applied.
5. Can I use **kubectl apply** for all resource types?
 - Yes, it works for **Deployments, Services, ConfigMaps, Secrets, Ingress**, and more.

Conclusion

kubectl apply is the **preferred method** for managing Kubernetes workloads. It enables **continuous deployment, easy updates, and infrastructure automation**, making it essential for **efficient Kubernetes operations**.